

# Additive Manufacturing of Flexible Strain Sensors Based on Smart Composites for Structural Health Monitoring with High Accuracy and Fidelity

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This research introduces a novel flexible spherical carbon nanoparticle-based polyurethane conductive ink, which is employed to fabricate strain sensors by a lab-developed direct ink writing/3D printing system. Rheological tests are performed, and sensors are pasted on glass fiber-reinforced plastic specimens to study strain gauge behaviors under quasistatic loading. The gauge factor in tensile loading is found to be layer width dependent as decreasing the strain gauge's layer width increases the sensitivity of the strain sensor. A maximum gauge factor of 34 is achieved using a layer width of 0.2 mm, 17 times greater than commercially available metal foil strain gauges. The four-point bend tests are performed under tension/compression to assess the sensor's strain-sensing and damage-monitoring ability. Fractographic analysis is coupled with strain monitoring using the developed sensor, which confirms that the failure progresses from intralaminar failure modes such as ply splitting in tension. At the same time, delamination leads to kink band formation under compression and the eventual failure of load-bearing fibers. The developed sensor exhibits repeatable performance with low hysteresis and integrated nonlinearity errors for up to 1000 cycles. The developed sensors could be effectively employed for online in situ structural health monitoring of aerospace structures under static and dynamic loading.

may still remain undetected.<sup>[3–6]</sup> In addition to the direct cost of inspection and repair (materials, labor, etc.), the indirect costs related to the withdrawal of the aircraft from operation have a greater economic effect on the user.<sup>[7]</sup> Many conventional nondestructive testing (NDT) methodologies<sup>[8]</sup> such as X-ray tomography,<sup>[9]</sup> eddy current analysis,<sup>[10]</sup> ultrasound monitoring,<sup>[11]</sup> etc. are used for noninvasive inspection of aircrafts.<sup>[12]</sup>

Structural health monitoring (SHM) is fast emerging as an alternative paradigm. Unlike traditional NDT systems, a distinguishing feature of SHM is that the system's sensors and actuators are permanently attached to the device, gathering whole field data almost continuously for the entire service life of the structure, without operator interference.<sup>[12–16]</sup> NDT is used to detect discontinuities in solids that are recognized as damage, either on the surface or inside.<sup>[17]</sup> On the other hand, SHM senses local changes in the structure, either in the properties of the material or its functionality.<sup>[12,16,18]</sup> These changes are detected by

comparing the structure's response to a stimulus with that of the pristine structure's response.<sup>[19,20]</sup> In terms of its implementation and design, SHM is a multidisciplinary domain that employs different sensors, including strain gauges,<sup>[21,22]</sup> acoustic probes,<sup>[23]</sup> interferometry, and fiber Bragg grating (FBG) sensors.<sup>[24]</sup> Aerospace structures use strain gauges for timely damage detection, monitoring crack propagation and averting catastrophic failures. By incorporating SHM system in an aircraft structure,

## 1. Introduction

Owing to the sophistication of aircraft structures, detailed disassembly and reassembly is required for inspection. Due to the complexity of its structure, certain critical sections of the aircraft are not readily accessible for inspection.<sup>[1,2]</sup> Even though maintenance in aerospace structures needs excessive amount of man power and man hours, some of the hidden subsurface cracks

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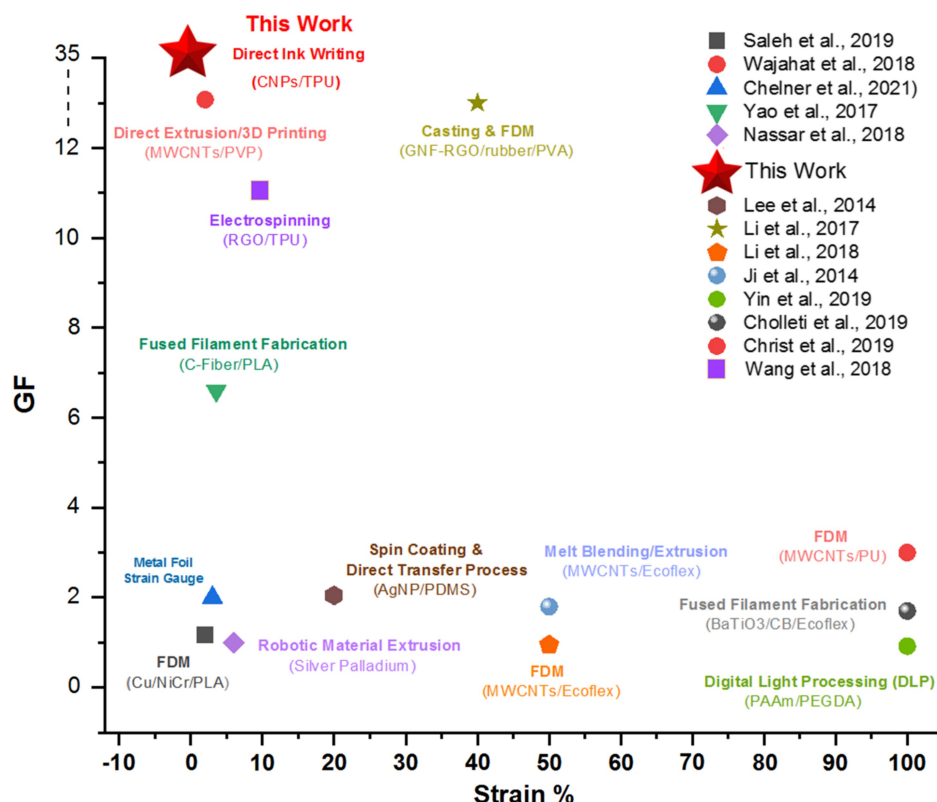
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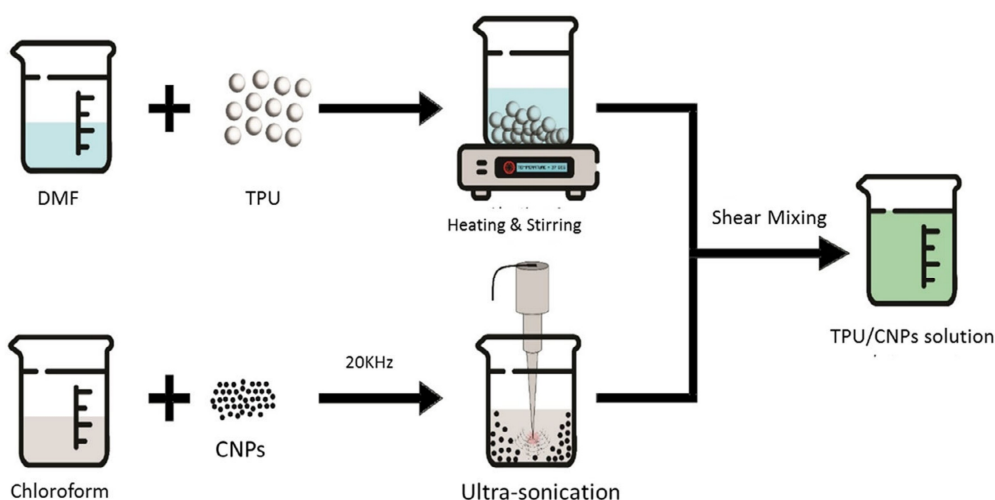
designer would not have to overbuild the wing structure to stiffen it which adds weight to the aircraft.<sup>[25]</sup> With a reliable and sensitive sensor, designer would have the freedom to create new damage tolerant designs for hypersonic aircraft and spacecraft.

Flexible polymeric sensors offer significant advantages as compared to metal foil strain sensors, such as high sensitivity, high strain-measuring capability, elevated elongation at break, and ease of application on complex structures.<sup>[26–29]</sup> Several polymeric composite materials are used to develop strain

sensors, such as thermoplastic polyurethane (TPU), polystyrene (PS), thermoplastic polyethylene (TPE), polydimethylsiloxane (PDMS), silicon, rubbers, etc. as matrix material.<sup>[30–32]</sup> At the same time, graphene, carbon nanotubes, and silver nanoparticles have been widely used as conductive fillers to develop conductive percolation networks.<sup>[33–46]</sup> Various researchers have reported strain sensors fabricated through fused deposition modeling (FDM)/3D printing, electrospinning technologies, etc.<sup>[47–52]</sup>



**Figure 1.** A comparison between already developed sensors with the current work in terms of GF.



**Figure 2.** Schematic showing CNPs/DMF dispersion and TPU/DMF CPC solution.